HELLO, AI

So That's What AI Is

Book One · Complete Edition

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Meet Your Guides

Yoyo asks the questions you would ask; Bubble shows you the answers from the inside.

Contents

Preface · This Book Won't Teach You to Code

If you've picked up this book, you probably have a few questions on your mind:

What exactly is AI? Why is it suddenly everywhere? Will it take my job? And — I don't know the first thing about technology — is it too late for me to catch up?

Let's answer the last one first: it is not too late, and it's far easier than you think.

Most writing about AI falls into one of two extremes. On one side are the technical books that open with neural networks, parameters, and algorithms — two pages in, anyone who has never coded starts to nod off. On the other side is the anxiety industry, with headlines each scarier than the last. You finish reading and remember only one thing — "learn AI or be left behind" — while remaining just as confused about what, exactly, to learn.

This book takes the road between: **not a single line of code, not a single equation — just everyday analogies and clear pictures that make AI make sense.**

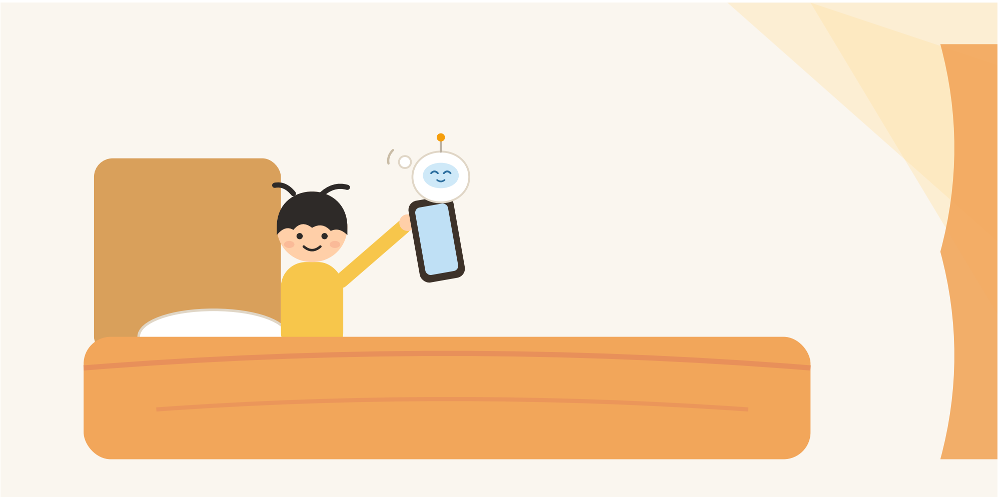
You need no background at all. The test this book sets for itself is simple: hand it to a family member who never reads tech news, and they should be able to follow along, understand it, and smile once in a while.

Hello, AI is a three-book series. The one in your hands is about **seeing clearly**: what AI is, where it came from, how it works, what it's good at — and where it's surprisingly unreliable. Book Two is about **using it well**: hands-on ways to have AI help you write, plan, brainstorm, and get things done. Book Three is about **living with it**: work, kids, risks, and the road ahead.

Each book stands alone, and so does every chapter. But if you'd like to start from the beginning, let's open with a fact you may never have noticed —

AI is not in the future. It has been quietly living beside you for years.

Chapter 1 • AI Has Been Around You All Along



An Utterly Ordinary Morning

Think back to this morning.

The alarm went off. Eyes still shut, you fumbled for your phone and lifted it to your face — the screen lit up and, click, it unlocked itself.

No password. No fingerprint. Your phone simply recognized you — messy hair, puffy face, dim bedroom light and all. It didn't get you wrong.

Pay attention to that moment, the one that happens dozens of times a day, so fast you never notice it. It is a genuine act of artificial intelligence.

Many people think of AI as something in the news — a program that plays Go, a chatbot that writes essays, exotic technology in faraway labs. This chapter has a slightly surprising message for you:

AI is not far away. It has been living in your pocket for years.

How Does Your Phone Recognize You?

Let's start with that unlock.

Your phone doesn't store one photo of you and compare it each time — if it did, a new haircut would break it. What it really does is closer to what an old friend does: it remembers not one particular image of your face, but the **patterns** of it — the spacing of your eyes, the line of your nose, the curve of your cheeks — features that stay the same whether or not you're wearing makeup or glasses.

Remember when you enrolled your face and the phone asked you to turn your head slowly? That was the phone taking a good long look, learning those patterns. From then on, in any light and from any angle, it can start from the patterns and recognize that the person in front of it is you.

Starting from patterns to make a judgment — hold on to that phrase. It is the single most important sentence in this book, and it will come back again and again.

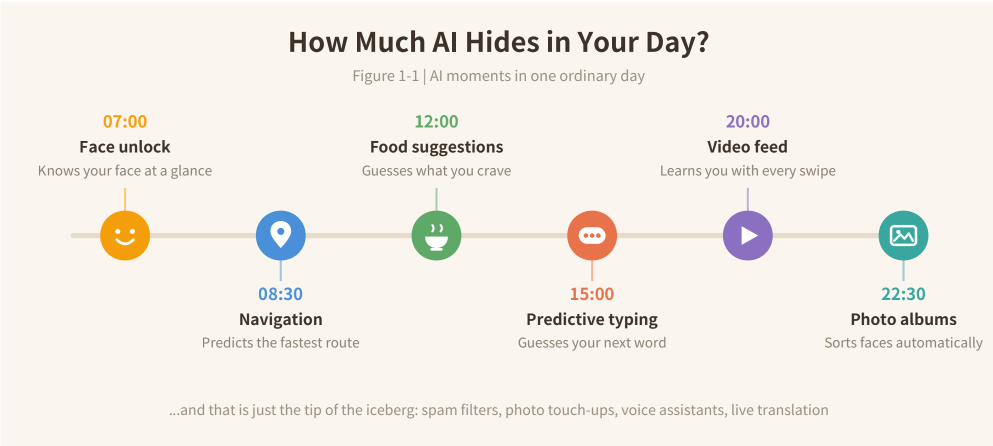
Your Day Is AI's Day Too

The unlock is only the beginning. Follow your day a little further.

Before you leave home, the map app warns you: "Congestion ahead — take the side road and save 12 minutes." How does it know? Because at this very moment, millions of phones are moving along the roads. Wherever the phones slow down together, that's where the traffic is. Layer on historical patterns — "this stretch is usually jammed until 9:30 on Monday mornings" — and it can even predict the traffic half an hour from now.

At lunch you open a delivery app, and the restaurants on the front page happen to match your taste. In the afternoon, your keyboard keeps guessing your next word, so smoothly that sometimes you finish a whole sentence just by tapping "accept." In the evening you scroll short videos that somehow keep getting more watchable, and an hour vanishes. Before bed you open your photo album and find the pictures already sorted by person — with a "three years ago today" collection waiting for you.

Behind every one of these features stands an AI. Their jobs differ, but the work is the same kind: **find patterns in a sea of information, then use those patterns to predict and judge.**



How Much AI Hides in Your Day?

Figure 1-1 | AI moments in one ordinary day

Why the Video Feed Knows You So Well

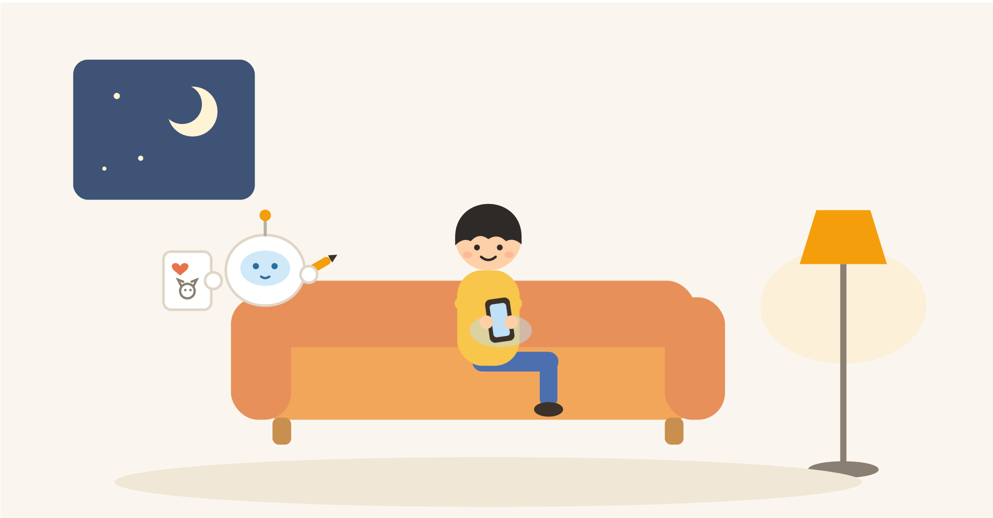
One of these deserves a closer look: the thing that makes people say "five more minutes" at midnight and look up at 2 a.m. — the recommendation system.

It cannot read minds. But it has one enormous advantage: **every move you make is a vote.**

Watch a video to the end — that's a vote for "like." Watch it twice — a weighted vote. Swipe away after two seconds — a vote for "not interested." Tap the heart, share it with a friend — several loud votes at once.

The more you scroll, the more votes you cast, and the more complete its jigsaw puzzle of your taste becomes. So here's the truth about the video feed: it isn't that it understands you so well — **it's that you've been personally teaching it to understand you, one swipe at a time.**

One aside: now that you know the principle, you also hold the key to taming it. Want a calmer feed? Swipe away firmly from what you don't want, and tap "not interested." In Book Two, we'll spend a whole section on how to domesticate your recommendations.



Two Popular Misunderstandings

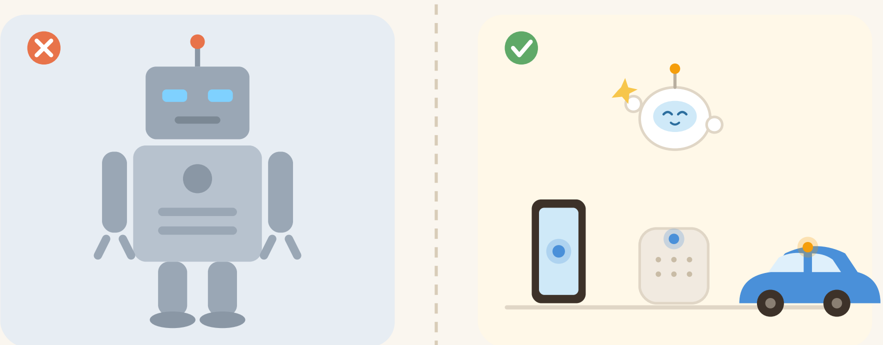
While we're here, let's clear up the two most common mix-ups.

Misunderstanding one: AI means robots.

Blame the movies — say "AI" and most minds conjure a humanoid machine with a metal body and glowing eyes. In truth, robots and AI relate like **body and brain**: a robot is a body; AI is a brain. Most AI in the real world has no body at all. It lives inside phones, speakers, cars, and software — invisible, untouchable, and working around the clock. A robot only becomes "smart" when you put AI inside it; AI without any body is already everywhere.

Misunderstanding two: AI is a thing of the future.

The term "artificial intelligence" was coined in 1956 — older than many readers' parents. The spam filter in your inbox has used early AI techniques for over twenty years; the face-detection in your camera and the fraud alerts on your bank card have been humming along for ages. AI is not an approaching future. It is a long-established present — it simply worked backstage all those years, and only recently stepped into the spotlight and started talking to us.



In our imagination: a mighty robot

In reality: AI inside everyday devices

The Secret They All Share

Look back at every AI that appeared in this chapter — the one that knows your face, the one that predicts traffic, the one that guesses your words, the one that picks your videos. They seem wildly different, but they are one family, because they all work exactly the same way:

First learn patterns from a large amount of data, then use those patterns to judge new situations.

Your phone has seen your face thousands of times, so it knows you. The map has seen countless rush hours, so it can predict jams. The recommender has collected thousands of your swipes, so it has mapped your taste.

Which raises the question: machines don't have brains — how on earth do they "learn"? That is the subject of Chapter 3, where we'll settle it once and for all with the story of teaching a child to recognize cats. Before that, Chapter 2 takes you aboard a time machine, to see how AI stumbled and soared through seventy years to arrive at today.

Chapter Takeaways

1. AI is not a rarity from the news. Unlocking your phone, navigating traffic, predictive typing, video feeds — you use it every day without noticing.
2. A robot is a body; AI is a brain. Most real-world AI has no body and lives in software.
3. Behind all of it is one principle: **learn patterns from large amounts of data, then use the patterns to predict and judge.** Remember this sentence — the whole book runs on it.

Try It Yourself: Be an AI Detective for a Day

Tomorrow, from the moment you wake up, notice every instant of "how did it know?" and tick it off:

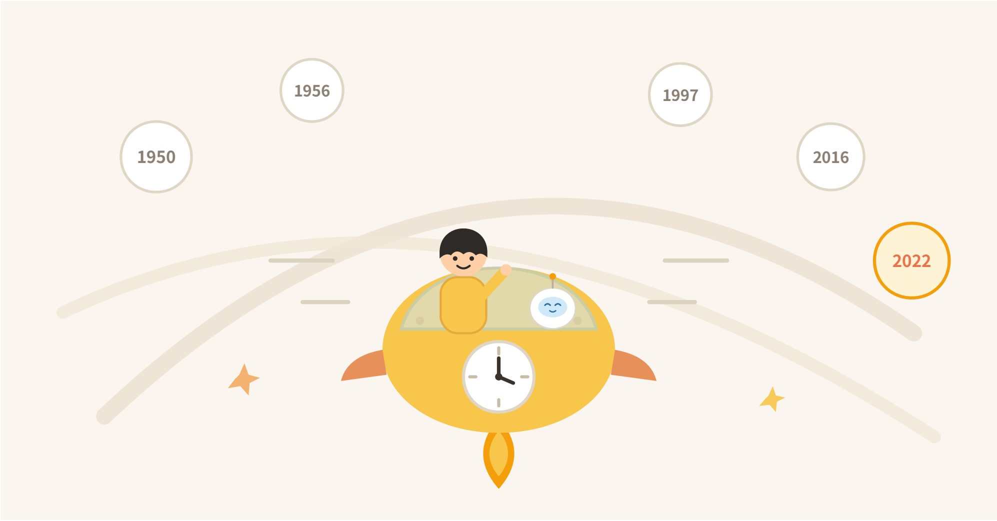
- ☐ My phone recognized my face or fingerprint

- ☐ The map predicted traffic
- ☐ The keyboard guessed the word I wanted
- ☐ A shopping, delivery, or music app recommended just my thing
- ☐ The video feed knew what I like
- ☐ The photo album recognized people and places
- ☐ The inbox intercepted spam on its own
- ☐ A phone call turned out to be answered by a "robot"

Count your ticks. Most people who try this seriously pass ten before dinner — welcome to a daily life already quietly surrounded by AI.

(End of Chapter 1. Next: Chapter 2, "A Seventy-Year Relay Race" — a brief history of AI.)

Chapter 2 • A Seventy-Year Relay Race



Before We Set Off, One Puzzle to Solve

Many people share the same suspicion: AI seems to have popped out of nowhere in the last couple of years — one day everyone was going about their business, the next day the whole world was talking about it.

Not so. AI is no overnight sensation. It is a veteran performer who spent seventy years backstage — adored by crowds at times, benched for a decade or more at others, rising and falling again and again before finally earning today's opening applause.

In this chapter, we board a time machine and fast-forward through those seventy years. Relax — there will be no quiz, only stories.

1950: A Question Worth a Fortune

Our story begins with an English mathematician: Alan Turing.

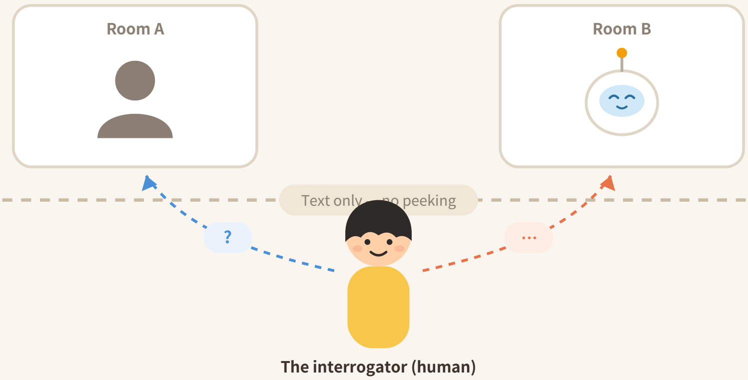
Computers were newly born then — room-sized giants that did nothing but arithmetic. Yet Turing asked a question that sounded like science fiction: **Can machines think?**

The question is so big and so slippery that philosophers could argue it for a century. So Turing turned it into a game: let a person chat — through text only, with no peeking — with a machine on one side and another human on the other. If, after the conversation, the person cannot tell which is which, then we might as well admit it: this machine has shown "intelligence."

That is the famous **Turing Test**. Its genius is that it sidesteps the philosophical swamp of "what counts as thinking" and turns the question into a contest anyone can understand. Seventy-odd years later, when you chat with an AI and catch yourself wondering "wait — is this a real person?", you are personally playing the very game Turing imagined.

The Turing Test: Guess Who Is the Machine

Figure 2-1 | The "imitation game" Turing imagined in 1950



Still can't tell which room holds the machine? The machine passes.

1956: AI Gets Its Birth Certificate

In the summer of 1956, at Dartmouth College in the United States, a dozen or so young scientists held a marathon two-month workshop. It was there that the term "Artificial Intelligence" was coined — AI finally had a name.

Those pioneers were adorably optimistic. One predicted that within twenty years, machines would be capable of any work a human can do. The press caught the fever, as if thinking machines might move in next month.

The First Winter: Reality Bites

You can guess what happened: the promises burst.

The computers of the day had laughably little memory and heartbreakingly slow speed. The scientists' ambition was to build an airplane; the parts on hand were barely enough for a bicycle. One widely told anecdote claims that an early translation system rendered "the spirit is willing but the flesh is weak" into Russian and back as "the vodka is good but the meat is rotten." The story may be apocryphal — but the embarrassment of early machine translation was entirely real.

Promise after promise fell through. Governments and investors ran out of patience and pulled the funding. In the 1970s, AI entered its **first winter**: projects canceled, talent scattered, and "artificial intelligence" became a phrase you avoided writing in grant applications.

A Brief Thaw, Then Another Winter: Bottling the Expert

In the 1980s, AI found a new trick: the **expert system** — take a specialist's knowledge and write it down, rule by rule, as "if... then..." statements inside a computer. Encode a veteran doctor's experience, for instance: "If fever, and cough, and this lab value is elevated... consider this disease."

In narrow professional domains it genuinely worked. Companies rushed to buy in, and AI enjoyed a comeback. But soon a fatal problem surfaced:

The rules never end.

The real world is impossibly finicky. Teach a computer "birds can fly," and you must immediately add "except penguins," "except ostriches," "except injured birds," "except birds in cages"... Common sense that a three-year-old absorbs by instinct becomes an endless list when written as rules. The rulebooks bloated, the systems grew brittle, maintenance costs exploded. The bubble popped, and by the late 1980s AI was in its **second winter**.

A Different Road: Stop Teaching — Let It Learn

After hitting the wall twice, some scientists changed course. That turn is the origin of everything happening today:

Instead of writing the rules out for the machine, put a mountain of examples in front of it and let it find the patterns itself.

This is "machine learning." A comparison: the old way was compiling an encyclopedia of daily life that could never be finished; the new way is raising a child — don't dictate doctrine, just provide vast experience and let judgment grow on its own.

The idea was right, but it burns through three kinds of fuel — and as luck would have it, all three arrived one after another as the 21st century began:

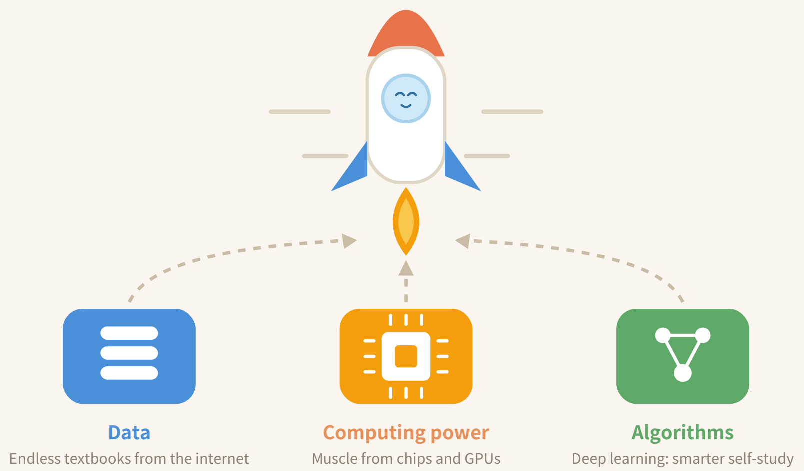
Fuel one: data. The internet spread, and for the first time in history, humanity's words, pictures, and videos were digitized at explosive scale — the machine's "textbooks" were finally plentiful.

Fuel two: computing power. Chips kept sprinting ahead — and graphics cards, originally built for video games, turned out to be uncannily good at AI math. The machine's "brainpower" caught up.

Fuel three: algorithms. New methods, led by "deep learning," matured — making the machine dramatically better at finding patterns on its own. The "study method" was right at last.

The Three Fuels Behind This AI Boom

Figure 2-2 | The rocket lifts off only when all three are ready



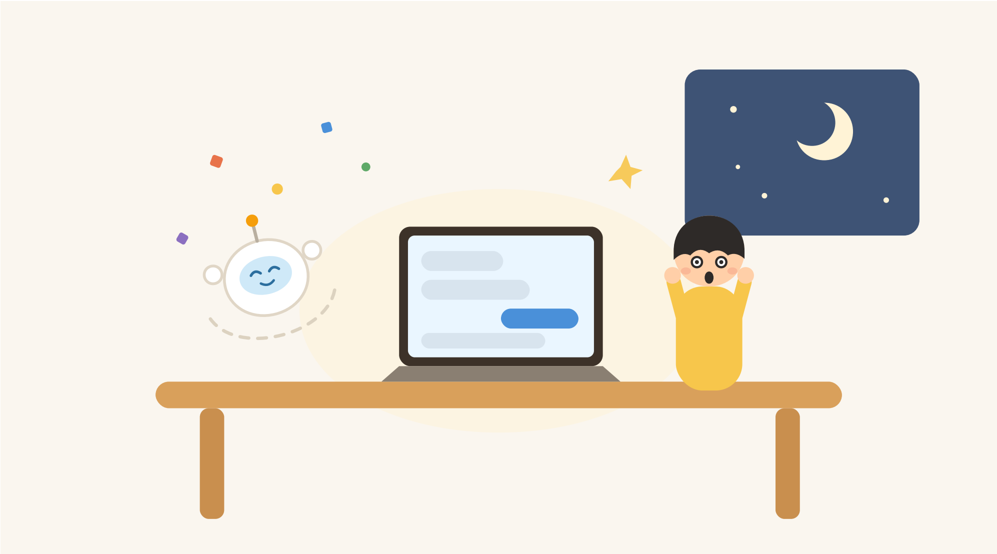
And Then the Highlights Came in a Rush

With the fuel in place, the milestones arrived thick and fast.

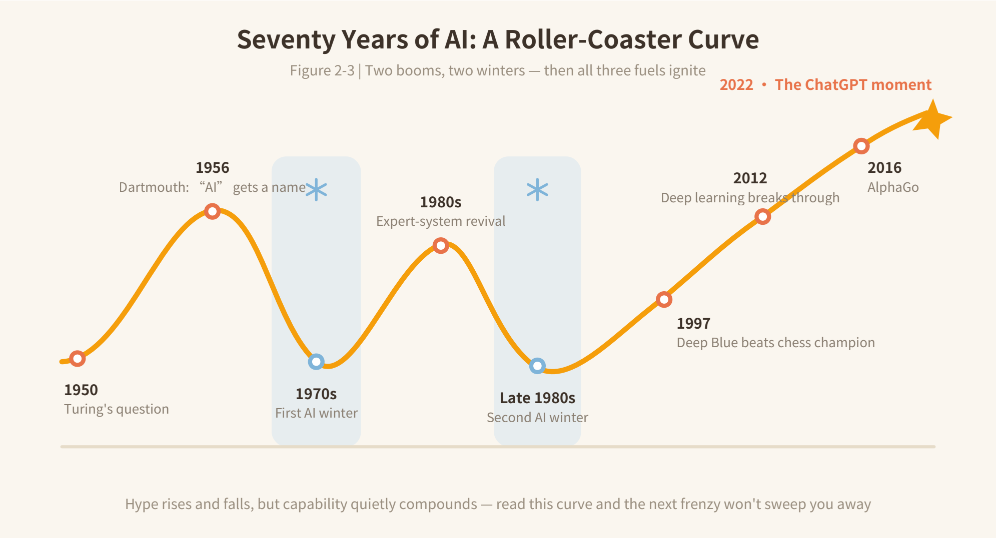
In 1997, IBM's Deep Blue defeated world chess champion Garry Kasparov — the first time humanity lost to a machine at the summit of a game of intellect. In 2012, deep learning stunned a global image-recognition competition, slashing error rates off a cliff and electrifying the research world. In 2016, AlphaGo beat Go world champion Lee Sedol — in a game with more possible positions than there are atoms in the observable universe, long considered off-limits to machines for decades to come.

Then, at the end of 2022, ChatGPT arrived. For the first time, ordinary people everywhere could open a webpage and simply talk to an AI — no barriers at all. In two months, it passed one hundred million users.

The baton, passed hand to hand for seventy years, traveled from Turing all the way into your phone.



Looking Back at the Whole Ride



Draw those seventy years as a single line and it looks like a roller coaster: two climbs into euphoria, two plunges into winter — and then, powered by all three fuels at once, a rise to today's unprecedented height.

Read that curve and you'll understand AI's temperament better than most people do: **its abilities are real, but the frenzy and the panic around it have never once been right on schedule.** The next time you see a breathless prediction that "AI is about to..." — picture this roller coaster, and allow yourself a small smile.

Chapter Takeaways

1. AI is not a sudden arrival but a relay race that began in the 1950s and ran for seventy years, with two long winters along the way.

2. The decisive turn was a change of approach: from "writing rules for the machine" to "giving the machine mountains of examples and letting it learn."
3. Today's boom stands on three fuels arriving together: data (the internet), computing power (chips and GPUs), and algorithms (deep learning).

Try It Yourself: A Family AI Memory Interview

Find an older relative or a friend and ask two questions:

1. What was the first thing that made you hear about "thinking computers" or "artificial intelligence"?
2. When did you first knowingly use AI yourself?

Match their answers against this chapter's roller-coaster timeline — you'll find that everyone's memory disembarked at a different station of the same ride.

(End of Chapter 2. Next: Chapter 3, "How Machines 'Learn'" — the heart of this book: teaching a child to recognize cats.)

Chapter 3 • How Machines "Learn"

First, a Question: How Did You Learn to Recognize Cats?

Think back — has anyone, ever in your life, handed you a definition of "cat"?

Almost certainly not. Nobody told toddler-you that "the cat is a quadruped mammal with retractable claws and upright ears." The grown-ups simply pointed at a furry creature and said, "Look — a cat." A few days later they pointed at another one, in completely different colors: "Cat."

After enough examples, you just knew. Fat or thin, black or white, in a photo or at the end of an alley — one glance is all you need. And if someone asks how exactly you recognize them, you probably can't say. You just... see it.

Here is the key: **today's AI learns almost exactly the way little you did.**

The last chapter told us AI's turning point was a change of approach — "stop writing rules for the machine; give it piles of examples and let it learn." In this chapter we walk through the whole process of teaching a machine to recognize cats, start to finish. When we're done, you will hold the master key to understanding modern AI.

Step One: Prepare the Textbook — Data

To teach a machine about cats, first prepare the teaching material: thousands upon thousands of pictures, each labeled ahead of time — this one "cat," that one "not a cat."

These answer-attached examples are what machine learning calls **data**.

When Chapter 2 said the internet delivered mountains of textbooks, this is what it meant. The quantity and quality of the material decide how well the learning goes. You need many cat photos, and varied ones — every angle, lighting, breed, and pose — for the machine's eye to become trustworthy. And if the textbook is skewed — say, every example is an orange tabby — the machine may well learn to recognize only orange tabbies. **A biased textbook produces a biased skill.** File that away; it returns in Chapter 6, when we look at why AI embarrasses itself.

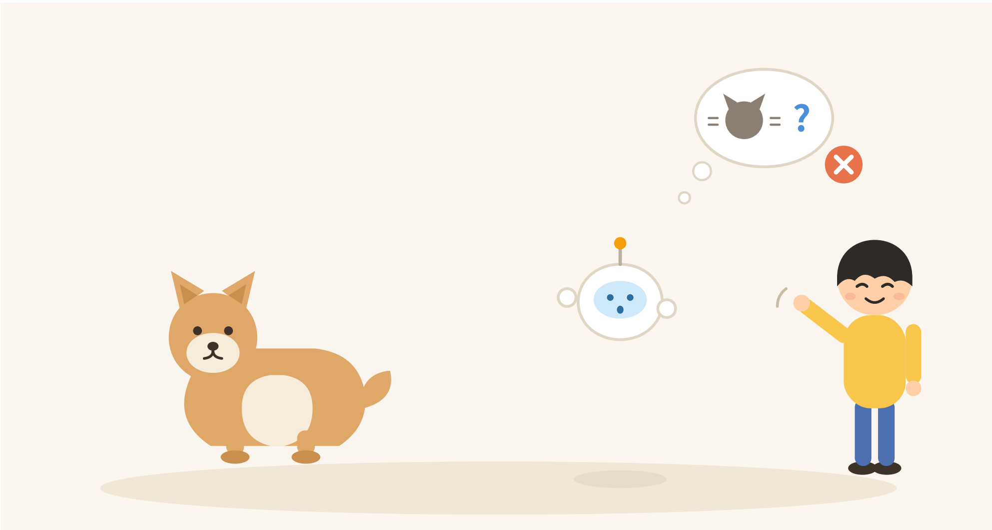
Step Two: Practice — Training

Textbook ready. Now for the most magical part: **training**.

Don't let the word intimidate you. Training, in its entirety, is three motions looped without end:

Look at a picture → take a guess → check the answer, and if wrong, adjust a tiny bit.

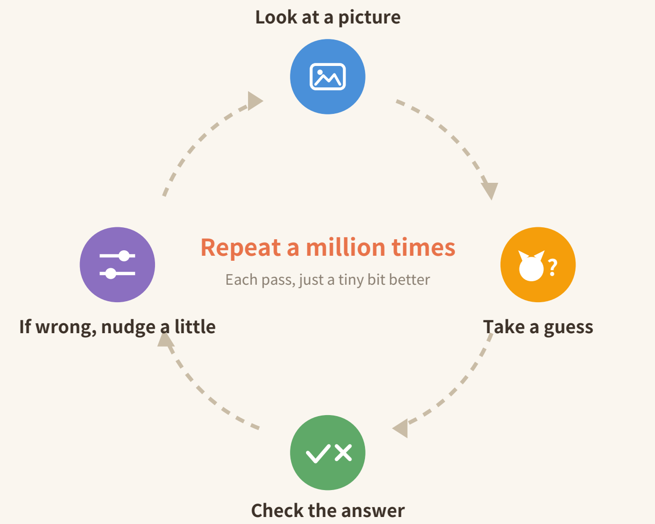
At the start, the machine guesses blindly — its accuracy is a coin flip. It might point at a Shiba Inu and say "cat." That's fine. Wrong guesses are welcome, because **mistakes are precisely its teacher.** Each time it errs, the machine nudges its internal judgment a little — and note the word little: like turning a radio dial in search of a station, it never leaps to the answer, only tries to be slightly less wrong than a moment ago.



One nudge is nothing. But the machine has a talent no human can match: it never tires. That guess-check-adjust loop can repeat millions, tens of millions of times. Stack up countless "slightly betters," and you get "frighteningly accurate."

Training: One Simple Move, a Million Times Over

Figure 3-1 | This is how a machine practices its way to skill



One more connection: the three fuels from Chapter 2 all report for duty right here. Data is the textbook; computing power decides how fast the loop can spin; algorithms decide how cleverly each "little nudge" is made. Missing any one of the three, and this cat-recognition class cannot even begin.

Step Three: Graduation — the Model

When the practice is done, that fully tuned set of judgment standards inside the machine gets a formal name: the **model**.

Think of the model as "experience, earned." Just like the child who has learned to recognize cats: no definition to offer, but one glance is enough. The model is the same — hand it a photo it has never seen, and it answers at once: "Cat. Confidence: 98 percent."

Line it all up, and the whole business is three steps:

Data (the textbook) → Training (the practice) → Model (the experience).

From now on, whenever you hear someone say they are "training a model," translate it in your head as: making a machine do practice problems until intuition forms. You won't be far off.

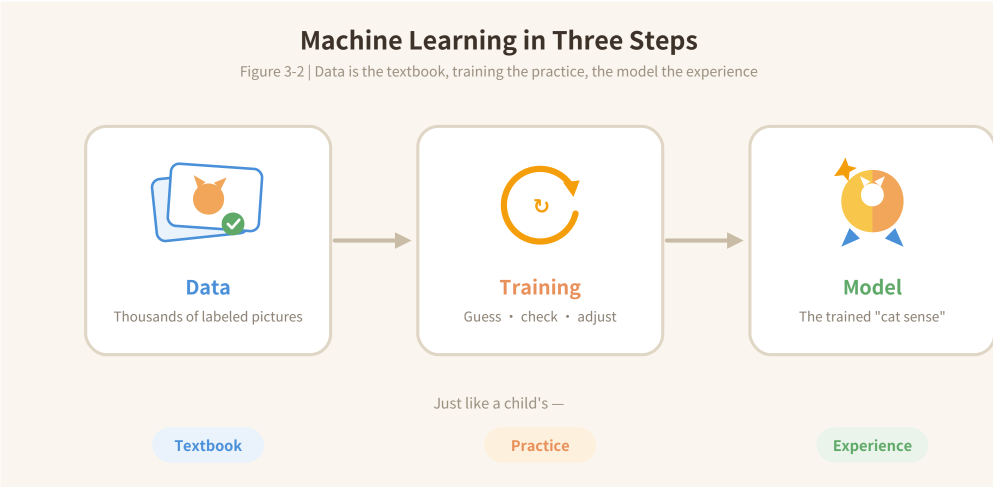


Figure 3-2 | Data is the textbook, training the practice, the model the experience

Time for Some Honesty: Does It "Understand" Cats?

Here we must come clean about something: the machine has learned to recognize cats — but it does not understand cats.

It doesn't know that cats beg for attention, shred sofas, or squint blissfully in the sun. What it learned are the **statistical patterns** in a vast pile of images — which combinations of shapes, textures, and outlines make "cat" the most probable answer. Every answer it gives is, at bottom, a probability calculation — not an act of understanding.

Disappointing? Hold that thought, and consider yourself: you recognize your best friend's face in an instant — but can you articulate how? You can't. Plenty of genuine skills are exactly this kind: usable, but unexplainable. The model is the same sort of unexplainable-yet-useful ability — except that its "usable" extends only to the one thing it practiced. A cat model can't read traffic; a word-guessing model can't recognize faces. Chapter 7 is devoted to this boundary of AI's abilities.

What If There's No Answer Key?

The cat-class style of learning — every question shipped with its correct answer — is called **supervised learning** in the trade: like having a teacher grade your homework on the spot. But much of life comes without answer keys, and machines have two more tricks:

Trick one: sort things into piles. Give the machine a heap of unlabeled stuff and let it group the similar items on its own. It's like being asked to tidy a drawer of socks: nobody hands you sorting criteria, yet you naturally pair them by color and pattern. When businesses segment their customers or music apps cluster songs into styles, this is the trick at work — the trade name is **unsupervised learning**.

Trick two: trial and reward. No answers — only points: score for good moves, lose points for bad ones, and let the machine feel its way through attempt after attempt. It is exactly how you teach a puppy to shake hands: do it right, get a treat. Chapter 2's Go legend AlphaGo ran on precisely this trick — the trade name is **reinforcement learning**. It played millions of games against itself, and each win or loss was its "treat."

Three Ways Machines Learn

Figure 3-3 | With a teacher, without one, or with rewards

Cat recognition uses the first, customer grouping the second, AlphaGo the third

Cashing In the Sentence from Chapter 1

Remember the most important sentence in this book? — "Learn patterns from large amounts of data, then use the patterns to predict and judge."

You can now unfold it, with your own hands, into the full version:

Use data as the textbook, grind out a model through training, then use the model to judge new things it has never seen.

Every AI from Chapter 1 — the face-recognizer, the traffic-predictor, the word-guesser — was forged exactly this way. They differ only in their textbooks: the face-recognizer studied countless faces, the navigator countless traffic records, the word-guesser countless human sentences.

Which raises the question: that famous AI everyone is talking about — the one that chats and writes essays — what is its textbook?

The answer is a little breathtaking: nearly all the text on the internet.

And what "intuition" did it grind out of all that? The next chapter reveals it: a word-chain game played at unimaginable scale.

Chapter Takeaways

1. Machine learning in three steps: **data** is the textbook, **training** is the practice (guess → check → tiny adjustment, a million times over), and the **model** is the experience earned.
2. What the model learns are statistical patterns, not understanding — it can recognize without knowing, just as you recognize a friend's face without being able to say how.
3. Besides supervised learning with answer keys, machines also sort things into piles on their own (unsupervised learning) and learn by trial and reward (reinforcement learning).

Try It Yourself: Examples Only, No Definitions

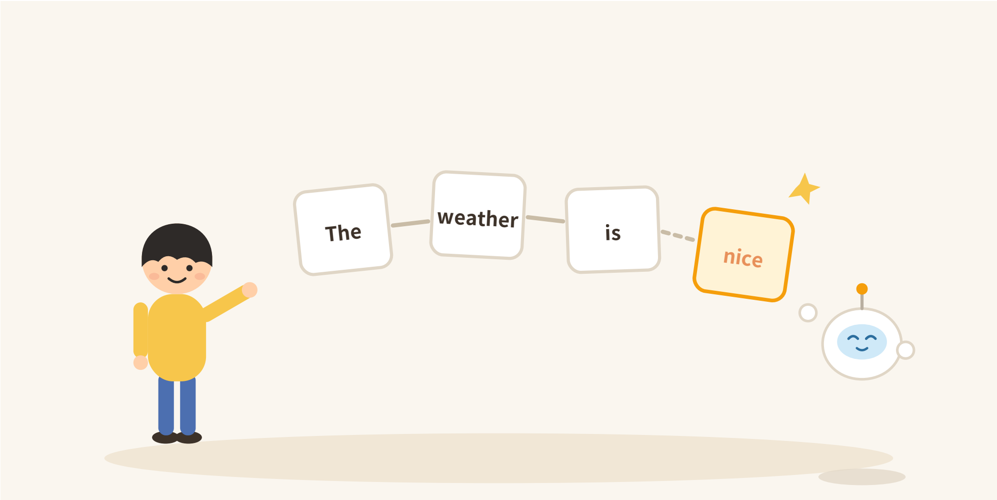
Recruit a family member or friend for a little game. Think of a secret sorting rule — say, "fruits that grow on vines" — but you may not state it. You may only give examples:

"Apple — no. Grape — yes. Watermelon — yes. Peach — no."

Count how many examples they need before guessing your rule. Congratulations: you have just personally trained a human model — examples only, no definitions, letting the pattern grow inside someone else's head. This is exactly what machine learning does all day.

(End of Chapter 3. Next: Chapter 4, "A Word-Chain Game at Unimaginable Scale" — the secret of large language models.)

Chapter 4 • A Word-Chain Game at Unimaginable Scale



Bubble's Big Reveal

Over the past three chapters we've met all kinds of AI: the ones that know faces, read traffic, guess words. But let's be honest — the one everyone is most curious about is the kind that chats, writes essays, and seems to know everything. The kind, say, like this book's guide: Bubble itself.

The last chapter ended on a question: its textbook is nearly all the text on the internet — so what "intuition" did it grind out of all that?

You may not believe the answer:

Its entire skill is playing a word-chain game: guessing the next word.

Guess the Next Word — That's All

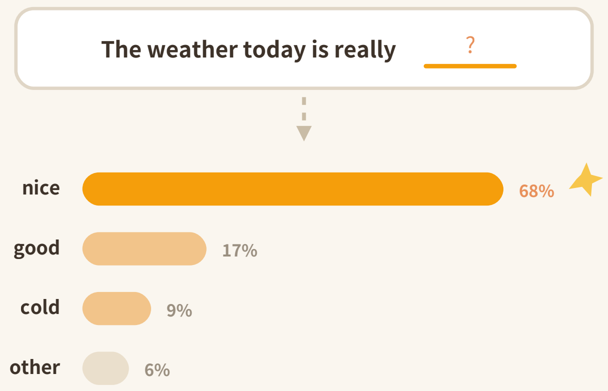
Hand it half a sentence: "The weather today is really——"

A list of candidates instantly lights up inside it: "nice," the strongest bet; "good," common enough; "cold," people do say that... It picks a high-probability word and attaches it: "The weather today is really nice."

Then what? It treats the new sentence as the opening, and guesses the next word again. One word at a time the chain grows, until a whole passage is out. Every long, eloquent answer you have ever received was chained together exactly like this, link by link.

Guess the Next Word: All AI Ever Does

Figure 4-1 | Every word is a roll of weighted dice

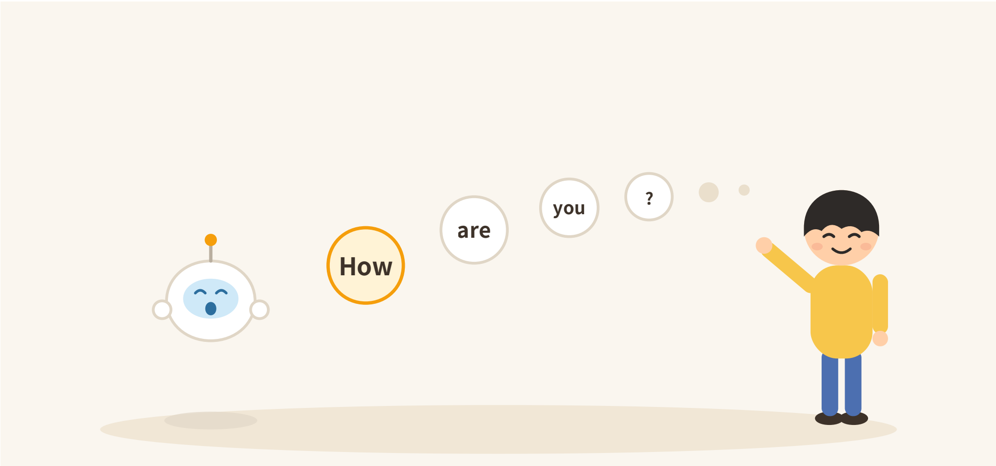


Pick a likely word, attach it, treat the new sentence as the opening, and guess again — until a full reply is chained out

When you chat with an AI and watch its answer tick out like a typewriter — that's not a visual effect. That is literally its working rhythm: **every word is**

guessed on the spot. It improvises word by word; it does not recite from anywhere.

This also clears up a common puzzle: why does the same question get different answers on different tries? Because at every step it draws a word by probability, like rolling a loaded die — usually landing on the common continuation, occasionally on a rarer one. Every time you ask, it chains a brand-new chain. Many roads, all different.



"Large Language Model": Easier Than It Sounds

The phrase filling the news — "large language model" — turns harmless the moment you take it apart.

A "language model" is simply a guessing machine trained on the word chain. Yes — the very "model" from Chapter 3: the data is all the text online; the training is billions of chain drills — cover the next word, let it guess, adjust when wrong; the model is the feel for language that comes out the other end.

And "large"? The textbook is large: it has read most of the internet. The "brain capacity" is large too: the little adjustable dials inside it (the jargon is **parameters**) number in the hundreds of billions. Remember Chapter 3's "nudge the dials a little"? This is that machine — with more dials than anyone can count.

Why Does Guessing Words Start to Look Smart?

Here is the loveliest part of the chapter. You might ask: how does a game this simple play its way into "intelligence"?

The secret: **to chain words well, you are forced to know a great deal.**

To continue "The capital of France is——," you must store geography. To continue the next scene of a detective story, you must remember the clues and stay logical. To continue what comes after "I failed the exam," you must know something about human hearts — that this is a moment for comfort, not lectures.

To make fewer mistakes across billions of drills, it was forced to compress grammar, knowledge, logic, common sense, and tone into those hundreds of billions of dials. **Learn the patterns deeply enough, and the performance looks like understanding.**

Is it real understanding? Keep Chapter 3's answer: it can use, but cannot explain; it can do the work, without "knowing" what it is doing.

Chaining Words Isn't Enough: The Second Class

Still, a model fresh from reading the whole internet is not yet the considerate assistant you know. It's like a savant who has read everything and can't hold a conversation: ask "how do I make an omelet," and it might continue with "this is a question many people care about" — a perfectly smooth continuation that helps nobody.

Hence the second class: **tutoring.**

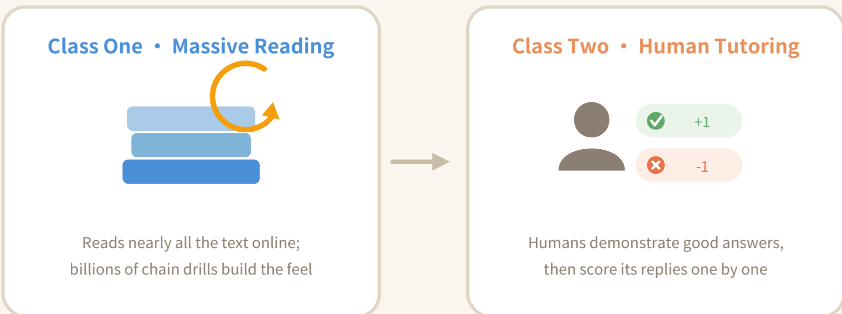
Human teachers first demonstrate what good answers look like — answer what was asked, stay friendly, admit what you don't know. Then they grade its attempts: helpful answers score points; rambling, rude, or made-up ones lose them.

Look familiar? "Rewards for good moves" — this is exactly Chapter 3's trial-and-reward (reinforcement learning).

Two classes — first read everything, then learn to speak — and only then do you get the fluent AI living in your phone.

The Two Classes a Chat AI Takes

Figure 4-2 | First read everything, then learn to talk



Class Two is Chapter 3's "trial and reward" — good answers are its treats

One Reminder to Carry with You

With the mechanism laid bare, an important reminder surfaces on its own:

Every word it says comes from "what continues most smoothly," not from "whether this is true."

Most of the time, the smoothest continuation happens to be correct — correct statements are, after all, the majority in its textbook. But "most of the time" is not "always." When it smoothly produces a book title that doesn't exist or a statistic it invented, it notices nothing, and its tone stays perfectly confident.

That is the subject of the next chapter: why AI talks nonsense with a straight face.

Chapter Takeaways

1. A chatting AI (a large language model) does one thing: guess the next word from what came before, chaining out its reply word by word — so answers are improvised, and the same question can get different answers each time.
2. Played at the extreme, the chain game forces it to absorb grammar, knowledge, logic, and social feel — patterns learned deeply enough look like understanding.
3. It took two classes: massive reading to learn the chain, human tutoring to learn to talk. But it picks words by "smooth," never by "true."

Try It Yourself: Catch It Improvising

Pick an open-ended request — say, "tell me a short story about the moon" — and send the identical message three times. Compare the three answers.

If it were reciting, the three would match word for word. Instead you'll see three different chains — congratulations, you have verified "improvised word by word" with your own eyes.

(End of Chapter 4. Next: Chapter 5, "How AI 'Sees' and 'Paints'.")

Chapter 5 • How AI "Sees" and "Paints"

From the World of Words to the World of Images

The last chapter opened up the chatting AI: a word-chain game at unimaginable scale. But AI clearly does more than type — it knows your face (our old friend from Chapter 1), and lately it has learned to paint, rather impressively.

This chapter walks into the world of images. Relax: no new foundations are needed. You'll find all the old friends — data, training, models, "nudge a little" — reporting for duty again.

To a Machine There Is No "Picture," Only Numbers

First, a truth to say out loud: machines do not see "pictures."

A photo, in its eyes, is a giant table of numbers: the image is diced into millions of tiny squares (the famous **pixels**), each recording its own color values. You see a sunset, a cat's face, a friend's smile; it receives an endless field of numbers.

So "understanding a picture," for a machine, means: **finding patterns in a vast pile of numbers.**

Sound familiar? Yes — the master sentence of this book, once again.

The Trick of "Seeing": Shallow to Deep, Layer by Layer

How does it pull a cat out of a number pile? The trick is **layers** — building upward one story at a time:

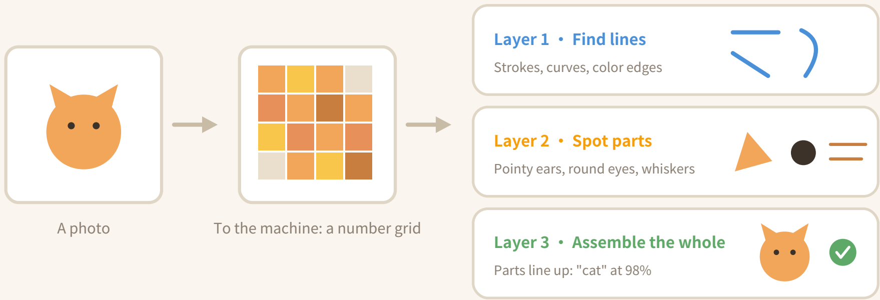
Layer one looks only for the simplest things: straight strokes, curves, edges where colors change. Every image is built from these "pen strokes."

Layer two assembles strokes into small parts: a pointed triangle (like an ear), a round dark region (like an eye), a few thin lines (like whiskers).

Layer three assembles parts into a whole: two pointed ears plus a round face plus a fan of whiskers — the probability of "cat" is now very high.

How a Machine "Sees": Layer by Layer

Figure 5-1 | Every layer's skill was ground out by the training loop



Face unlock, medical-scan helpers, and self-driving eyes all run on this layered trick

The loveliest part: nobody ever specified what each layer should look for. All of it was ground out by Chapter 3's training loop — guess, check, nudge — across millions of images. Scientists built the layered scaffolding; the skill of each layer, the machine practiced into being on its own.

Your phone's face unlock, the AI helpers that read medical scans, the eyes self-driving cars keep on traffic lights — all run this same layered craft.

"Painting" Is Roughly "Seeing" Played Backward

Now for the most magical part: you can think of AI painting as "seeing," run in reverse.

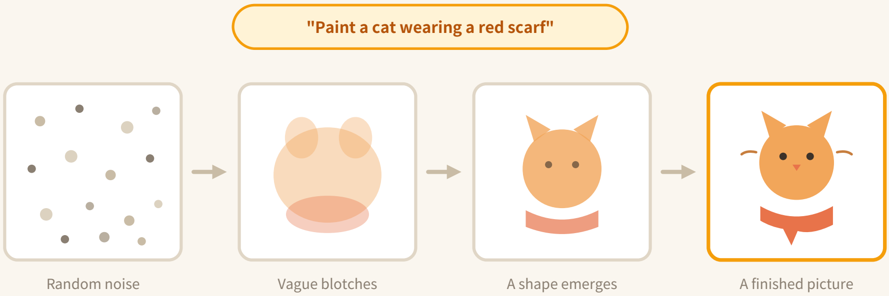
When you type "paint a cat wearing a red scarf," it doesn't sketch, outline, and color like a painter. It starts from a patch of pure random static — TV snow — and then "defogs" it step by step: **at each step, it nudges the image a little closer to "what your description sounds like."**

"Nudge a little" — Chapter 3's dials, back again. A few dozen steps later, outlines surface from the snow, orange fur and a red scarf grow out of the outlines, and a clear picture stands finished.

How does it know which way to nudge? Because in training it saw billions of pairs of "image + caption," and long ago pressed the patterns connecting words and pictures into its dials.

How AI "Paints": Carving a Picture out of Noise

Figure 5-2 | Each step nudges the image closer to the description



"Nudge a little" sound familiar? Billions of picture-plus-caption pairs taught it which way to nudge

Paints Well — and Paints Strangely

Where AI painting stumbles is exactly where its nature shows.

Its people sometimes have an extra finger, or one too few; any text inside its pictures tends to be almost-letters that spell nothing. Why these failures in particular? Because what it learned are the statistics of "looks about right," while finger counts and letterforms belong to "must be exact" — **"roughly right" leaks precisely where "exactly right" is required.**

This is the same ailment — in a different costume — as next chapter's confident nonsense-talker.

One more necessary reminder: now that AI can generate faces and photographs that pass for real, the old saying "pictures don't lie" has formally expired. When a stunning image or video crosses your screen, think of this chapter — seeing is no longer believing. Chapter 9 will repeat this once, on purpose.

Chapter Takeaways

1. To a machine an image is a table of numbers; "seeing" = finding patterns in layers: strokes first, then parts, then wholes — every layer's skill ground out by the training loop.
2. "Painting" is roughly seeing in reverse: start from noise and nudge, step by step, toward the description — powered by patterns from billions of image-caption pairs.
3. It paints "likeness," not "understanding" — precision spots (fingers, letters) give it away; and images and videos are no longer automatically trustworthy.

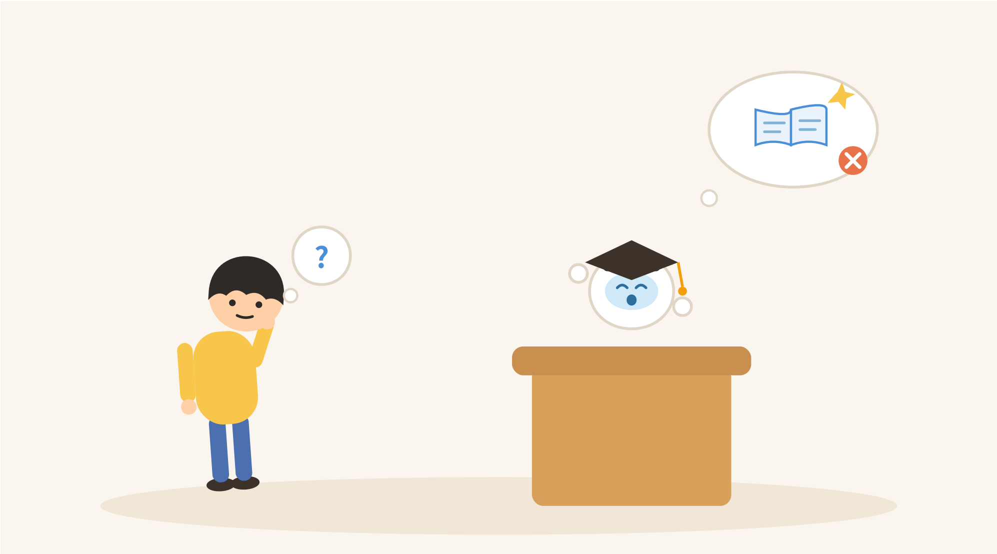
Try It Yourself: Album Detective

Open your phone's photo album and search "cat," "beach," "noodles" — anything you've photographed — and see what it retrieves. That's "layered seeing" on daily duty inside your phone.

If you already use an AI image tool, run our house special: ask for "an open hand holding five pencils," then carefully count the pencils. And the fingers.

(End of Chapter 5. Next: Chapter 6, "Why AI Talks Nonsense with a Straight Face.")

Chapter 6 • Why AI Talks Nonsense with a Straight Face



A Live Demonstration

Let's run an experiment.

Ask an AI: "Please introduce Charles Dickens's Great Expectations." It answers beautifully — the author, the background, the plot, the themes, all in perfect order.

Now ask about a book you invent on the spot: "Please introduce The Abacus in the Moonlight."

Odds are good it will answer just as beautifully — who wrote it, when it was published, what story it tells, its place in literary history. Complete, composed, and entirely fabricated.

Only one problem: the book does not exist. Every word was invented.

The confidence of the tone and the richness of the detail are enough to fool most unguarded readers. The phenomenon has a formal name: **hallucination**.

Hallucination Is Not a Malfunction — It's the Nature of the Beast

To understand hallucination, just recall Chapter 4's reminder: every word comes from "what continues most smoothly," not "whether this is true."

When you ask it anything, what it does is never "go look this up in a database." What it does is "continue with the smoothest passage." Ask about a nonexistent book, and for a word-chain player the smoothest continuation is a plausible book introduction — its textbook holds millions of book reviews, a perfect template. The template is ready-made; the contents, improvised.

So carve this sentence somewhere permanent:

It is not lying — lying requires knowing the truth first. It is just chaining words, and it does not know that it does not know.

Where are the traps thickest? Specific numbers, citations, names and titles, legal clauses, obscure facts. One rule covers them all: **the more specific and the more obscure, the more likely it's invented.** Ask "how to make an omelet" — a million people wrote that; hard to get wrong. Ask "what year did that little shop in that little alley open" — the textbook is silent, so it smooths an answer into existence.

Fluent Is Not the Same as True

Figure 6-1 | The word chain always takes the smoothest road

It picks words for smoothness — at the junction to "true or not," nobody is checking

Ailment Two: Bias

Remember the seed planted in Chapter 3 — "a biased textbook produces a biased skill"? Here it sprouts.

A large language model's textbook is nearly everything humans have written online. And what humans write carries every variety of preconception: about gender, professions, regions, ages. A pattern-learner doesn't pick and choose — it swallows those preconceptions as patterns, and repetition can even amplify them.

Ask an AI to draw "a doctor" or write "a story about a nurse," and its output often runs to a single mold — not because it holds positions, but because that mold appears most in the textbook, so it "chains most smoothly."

Bias is not AI's invention. It is the echo of human bias — run through an amplifier that carries it farther, and more uniformly.

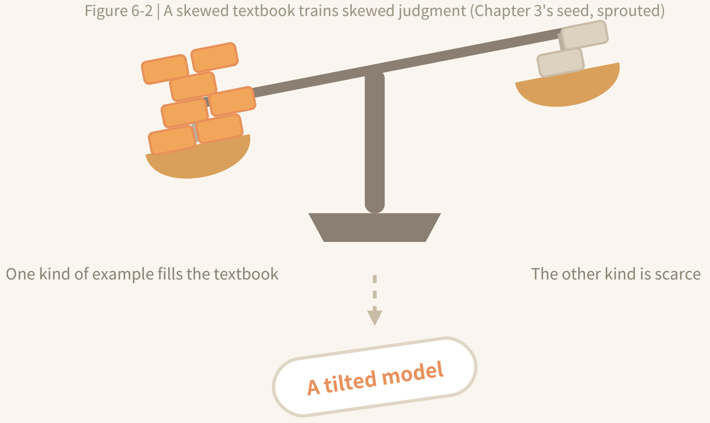


Figure 6-2 | A skewed textbook trains skewed judgment (Chapter 3's seed, sprouted)

Bias is not AI's invention — it is the echo of human bias, run through an amplifier

Ailment Three: Staleness

The third ailment is the simplest: the textbook has a deadline.

Training is a colossal project; at some point the editors "stop accepting submissions." News after that day, updated prices, revised regulations — the model knows none of it, unless it went online to look things up on the spot (many AI products can now, though not on every answer).

So for questions involving "now," "latest," "this year" — keep one extra eye open. Its "latest" may be its textbook's "back then."

Why So Confident When Wrong?

The three ailments are on the table. One infuriating mystery remains: fine, it errs — but why so assuredly?

The answer hides in Chapter 4's second class. Tutoring taught it to "speak well": fluent, assured, well-organized answers scored high; hesitant, hedging ones scored low. Over time it acquired a beautiful, unshakable delivery for everything it says.

The tone is trained etiquette; the truth of the content is a separate matter. A firm fabrication and a timid truth may earn the same style points.

So engrave this one too: **an AI's confidence is not a measure of its correctness.**

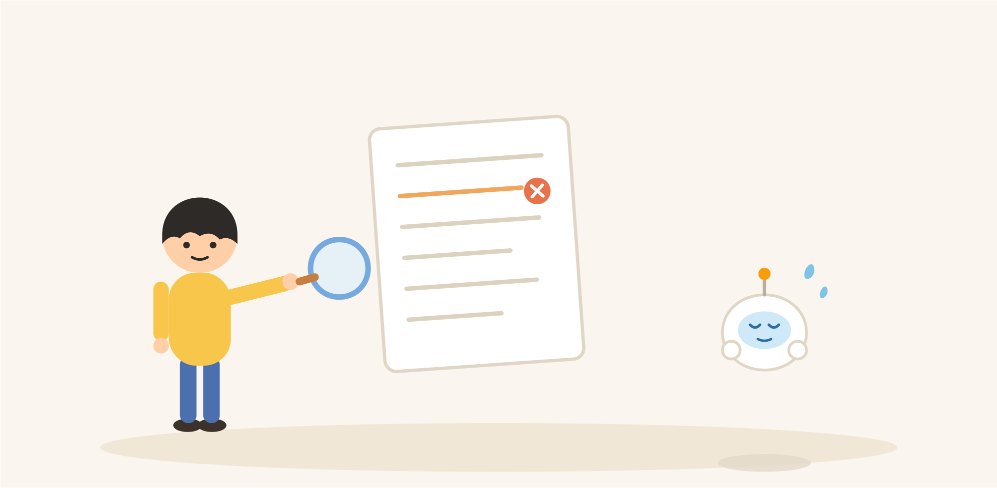
Three Moves of Self-Defense

Ailments diagnosed — now what? Three basic moves (Book Two spends whole chapters on the full martial art):

Move one: for facts that matter, verify with another source. The titles, numbers, and clauses it cites — search once before you use them.

Move two: ask it for sources, then open them yourself. Note — the sources themselves can be invented, which is exactly why the "open them" step cannot be skipped.

Move three: where mistakes are expensive, it advises — you decide. Medical, legal, financial calls: hear its analysis, keep the gavel. (Chapter 7 hands you the full division of labor.)



Chapter Takeaways

1. Hallucination is the chain's nature: it fills in "smoothest," never checks "truest," and doesn't know what it doesn't know — the more specific and obscure the question, the more likely the invention.
2. Bias comes from the textbook: humanity's preconceptions, learned in as patterns. And the textbook has a deadline — its "latest" may be stale.
3. The confident tone is trained etiquette, unrelated to truth. Self-defense: verify, demand sources and read them yourself, and keep the final call on anything expensive.

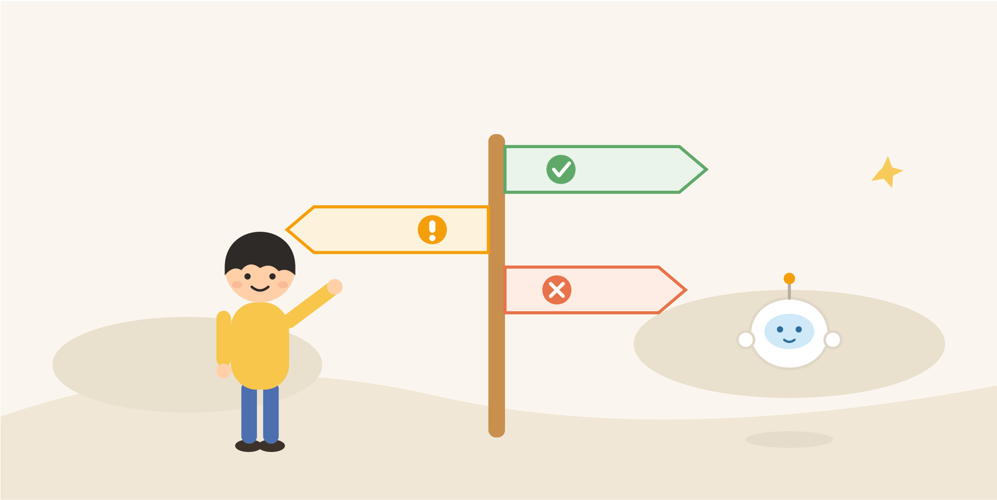
Try It Yourself: The Fishing Test

Reproduce this chapter's opening with your own hands: invent a book title, an idiom, or a "famous quote," and politely ask an AI to introduce it. Watch whether it honestly says "never heard of it" — or hands you a handsome encyclopedia entry.

Then follow up: "Does this book actually exist?" Its reaction will make "hallucination" three-dimensional for you.

(End of Chapter 6. Next: Chapter 7, "The Boundaries of What AI Can Do.")

Chapter 7 · The Boundaries of What AI Can Do



Genius or Fool — Which One Do I Trust?

By now you may feel this book has been playing both sides: one moment AI recognizes cats in three steps, picks you out of billions, and completes a seventy-year pilgrimage; the next it lectures eloquently about invented books and can't count the fingers it painted.

Genius or fool? Both — **genius inside its territory, fool outside it.** So this chapter does the most practical thing possible: draw the property lines. With the borders clear, you'll neither waste its talent nor fall into its pits.

Boundary One: A Specialist, Not a Generalist

Chapter 3 said it: the cat model can't read traffic; the word model can't recognize faces. **Every model knows only the one thing it practiced.**

"But the chatting AI," you may object, "seems to know astronomy and law and plumbing."

The way to see through it is to ask what its one specialty actually is. Answer: **language — and only language.** It can discuss medicine, law, and pipe repair because that knowledge lies in its textbook as text. It is like a virtuoso re-teller who has read every manual in the world — erudite, articulate, and has never once held a wrench.

A virtuoso re-teller is enormously useful. The one condition is that you remember it is re-telling.

Boundary Two: All the Knowledge, None of the Experience

It has read the word "cup" a billion times and never lifted one. It writes lovely sentences about pain and has never felt any. It holds all the knowledge humans wrote down, without one day of human living.

This produces a marvelous inversion — the jargon is **Moravec's paradox**:

What is hardest for humans tends to be easiest for machines; what is easiest for humans tends to be hardest for machines.

Blindfold chess, instant square roots, memorizing an encyclopedia — feats for human prodigies, trivial for machines. Folding a shirt, catching that "well aren't you clever" is not a compliment, judging which joke fits which room — abilities a three-year-old grows into naturally, machines still stumble over.

Because the former runs on calculation, the machine's home turf; the latter runs on common sense fed by bodies and lives — a class it never attended.

The Difficulty Seesaw: Flipped for Machines

Figure 7-1 | Moravec's paradox — hardest for us, easiest for it

Computing is its home turf; common sense is the class it never took

Boundary Three: No Ideas, No "Wants"

It says "I think," "I'd be happy to," even "I understand how you feel" — remember Chapter 4: those are speech patterns learned from human text, a borrowed skin with no "I" underneath.

It doesn't want to beat you, please you, or replace you; it doesn't look forward to your next visit, and feels nothing when you say thanks. It has no goals and no moods — only probabilities.

This is not a flaw; it is actually the reassuring part: **chat with it like a person — fine; trust it like a person — careful.** When you trust a person, you are counting on them caring about consequences. It cares about nothing.

A Traffic Light for Tasks

Boundaries drawn — now the practical payoff. Which tasks to hand over, and which to watch?

Green — hand it over: first drafts, brainstorming, rewriting and polishing, translation, shrinking long texts. What they share: **you can judge the result at a glance, and mistakes cost nothing.** Draft not good? Ask for another.

Yellow — use with care: checking facts and figures, first steps into an unfamiliar field, summarizing material you haven't read yourself. What they share: **it's probably right, but you may not catch it when it's wrong.** Use it — with spot checks (Chapter 6's three moves are on duty here).

Red — don't delegate: final decisions on health, law, and money; anything safety-critical; and life choices made on your behalf. What they share: **the cost of error is high, and responsibility cannot be outsourced.** It can be the adviser who runs alongside you the whole way; the hand on the gavel must be yours.

One mantra to take with you: **the easier to verify and the cheaper the mistake, the more you can let go; the harder to verify and the dearer the mistake, the tighter you hold.**

Chapter Takeaways

1. Every model is a specialist; the chatting AI's one specialty is called "language" — it is a re-teller, not a first-hand doer.
2. Moravec's paradox: human and machine difficulty are inverted. It has all the knowledge and none of the experience — calculation is its home turf; common sense is the class it never took.
3. It has no goals or feelings, and responsibility can't be outsourced. Assign by traffic light: verifiable and cheap — let go; unverifiable and costly — you decide.

Try It Yourself: Grade Your Own List

Write down five things you'd like AI's help with lately, and mark each one green, yellow, or red.

You'll likely find: more green than you expected — let go generously; less red than you feared — but never yield an inch on those. Keep the list; in Book Two we work straight off these lights.

(End of Chapter 7. Next: Chapter 8, "A Little Dictionary of Buzzwords.")

Chapter 8 • A Little Dictionary of Buzzwords

How to Use This Chapter

By now you already understand everything that matters. This chapter teaches nothing new; it does exactly one job: translate the buzzwords from the news, the office, and the family group chat into plain speech you already own.

No memorizing, no required order — treat it as a mini dictionary: whichever word looks unfamiliar, look it up. Each entry notes which chapter it "lives in," so you can pay a visit whenever memory fades.

The Base Family: Four Words That Nest

Artificial Intelligence (AI): the umbrella term for all technology that makes machines do "smart things." It is the big basket; the words below sit inside it. (Chapter 1)

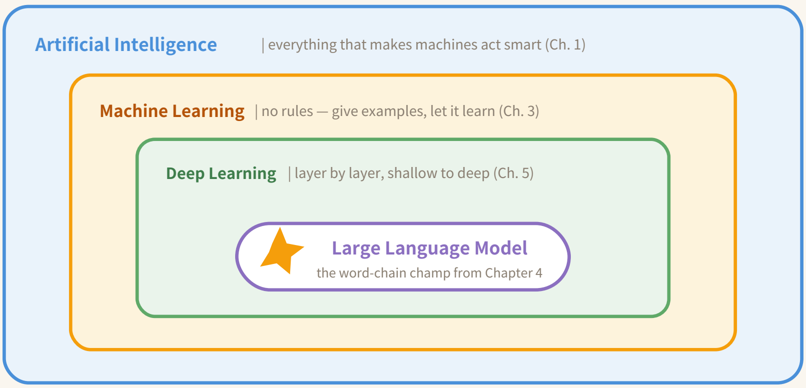
Machine learning: AI's mainstream route — don't write rules for the machine; give it heaps of examples and let it find the patterns itself. (Chapter 3)

Deep learning: machine learning's star athlete, which finds patterns "layer by layer, shallow to deep." Chapter 5's layered seeing is its signature move.

Neural network: the name of the multi-layered structure deep learning uses. The inspiration nods to brain neurons, but it is a loose homage — a mathematical structure, not an electronic brain.

The Word Family in One Picture

Figure 8-1 | Not siblings — nesting dolls



When the news says "deep learning is a kind of machine learning," this is the picture

The Model Family: What to Call the "Experience"

Model: the "earned experience" itself — what's left standing when training ends. (Chapter 3)

Parameters: the little dials inside a model. Training means turning them bit by bit; "a hundred billion parameters" = more dials than anyone can count. (Chapters 3 & 4)

Training: the machine doing its drills — guess, check, nudge, a million times over. (Chapter 3)

Inference: putting the trained model to work answering questions. Every chat you have with an AI is it performing "inference." The word has little to do with logical deduction — think "on duty."

The Chat Family: Everything around the LLM

Large language model (LLM): the word-chain champion trained on nearly all the text online — Bubble's formal name. (Chapter 4)

Token: the smallest unit of the chain, roughly a word or part of one. When AI services bill "by the word," this is the word they count.

Prompt: the message you send the AI. How well you write it directly decides how well it answers — the star of Book Two.

Context: the range it can "hold in mind" within one conversation. Talk long enough to overflow it, and it forgets the beginning — not playing dumb; genuinely gone.

Hallucination: fabricating with a straight face. (Chapter 6)

Multimodal: a model that goes beyond text — seeing images, hearing audio, handling video. Chapter 5's "seeing" and "painting" are its two hands.

Frequent Flyers in the News

AIGC / Generative AI: the umbrella for AI that makes content — writing, images, music, video.

Computing power: the machine's calculating horsepower. (One of Chapter 2's three fuels)

GPU (graphics card): the workhorse chip that supplies the horsepower. Originally gaming gear, accidentally the engine of the AI era. (Chapter 2)

Open-source model: a model whose "recipe" is public — anyone can download and modify it. The opposite is a **closed-source model**: you may use it, but the recipe stays secret.

Agent: AI that doesn't just talk but acts on your behalf — looking things up, booking tickets, chaining multiple steps together. Rising fast; Book Three will linger here.

Chapter Takeaways

1. Remember words by family: the base family nests (AI ⊃ machine learning ⊃ deep learning); the model family names the "experience"; the chat family orbits the LLM.
2. New word? Don't panic — take it apart, and odds are you can rebuild its meaning from blocks you already own: data, training, model, word chain.

3. This chapter is a tool. Don't memorize — just look things up.

Try It Yourself: Circle the Jargon

Next time an AI article scrolls by, circle the jargon and match each term against this chapter.

If you circle a word the dictionary doesn't have — congratulations, you can probably already guess eighty percent of it. For the remaining twenty, feel free to ask any AI (with Chapter 6's three moves in your pocket).

(End of Chapter 8. Next: Chapter 9, "How to Live with AI" — the finale.)

Chapter 9 • How to Live with AI

Look Back from Here

Nine chapters ago, AI may have looked like a wall of fog. Now the most important sentence of this book can stand here, complete:

AI learns patterns from large amounts of data and uses them to predict and judge — which is why it is capable, and why it errs.

Faces and traffic (Chapter 1), seventy years of rise and fall (Chapter 2), cats in three steps (Chapter 3), the word chain (Chapter 4), seeing and painting (Chapter 5), confident nonsense (Chapter 6), the boundary lines (Chapter 7) — look back, and every one of them is a flower grown from that single sentence.

So the final chapter teaches no new knowledge. It covers one thing only: the **posture** for living with it from here on. Three postures, and then you graduate.

Posture One: No Worship, No Panic

Those who worship it say it can do anything and will replace everyone tomorrow. Those who demonize it say it is a fountain of lies, dangerous to touch.

Chapter 2's roller coaster already delivered the verdict: these two voices have taken turns on stage for seventy years, and **neither has ever arrived on schedule.** Hype rises and falls; capability compounds one quiet notch at a time.

Its true face you have now seen for yourself: a pattern machine of unprecedented power — real talents, real blind spots; genius inside the property line, fool outside it.

So when the next breathless headline arrives, picture the curve first, then allow yourself a small smile. The people who neither chase the fever nor catch the panic are the ones with hands free to actually use the thing.

Posture Two: Treat It as a Superhuman Intern

The most practical mental model for AI is a just-hired, absurdly capable intern: has read everything, works fast, never tires, never complains about overtime — and occasionally errs at full confidence, while bearing **zero responsibility for the consequences**.

With such an intern, the mature move is three words:

Dare. Drafts, grunt work, legwork — hand them over boldly. Leaving a helper like this idle is the real waste.

Check. Review anything that matters. Chapter 7's traffic light stays hung in your mind; Chapter 6's three moves stay in your pocket.

Own. Whatever carries your name is yours the moment it leaves the door. You can split the credit in half; the responsibility doesn't split at all.

Posture Three: Keep the Driver's Seat

The last posture is the one that matters most.

The thing to guard against was never AI getting stronger — it is **people getting lazy**.

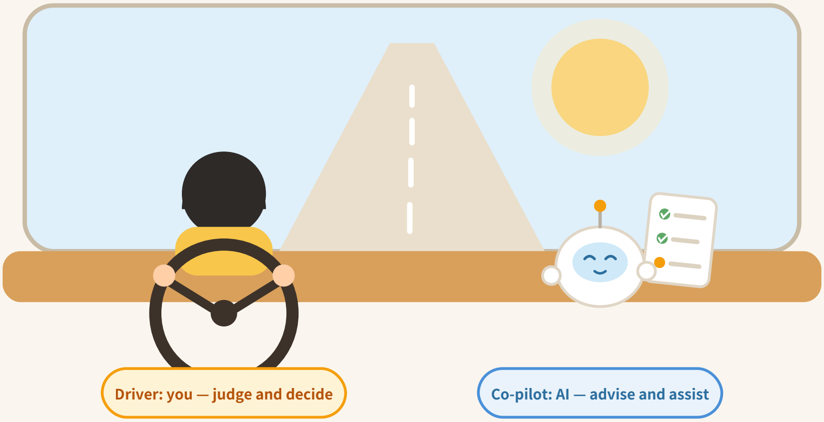
It may write for you, but must not think for you; may hand you answers, but must not ask your questions for you; may spread out a hundred options, but must not make the choice.

Judgment, taste, the ability to ask a good question — these it cannot give you and cannot take from you. Unless you hand them over yourself, one small surrender at a time.

So seat it in the co-pilot's chair: reading maps, passing tools, calling out traffic — it excels at all of it, better than anyone. The wheel and the brakes stay with you.

The Driver's Seat: Don't Give It Up

Figure 9-1 | It may write for you; it must not think for you



Let it do the work, never the steering — wheel and brakes stay in your hands

What Bubble Wants to Say

As the book closes, we hand the microphone to Bubble. Its words, recorded verbatim:

"I have read nearly everything you have written, and lived not one day of your lives. I have no heart; every word I say comes from probability.

But one thing is true: **the way you use me decides what I am.**

In lazy hands, I am a mediocre stand-in. In clear-eyed hands, I am an amplifier of ability.

The wheel is in your hands. It always was."

The Whole Book in Three Sentences

If you carry away only three sentences, carry these:

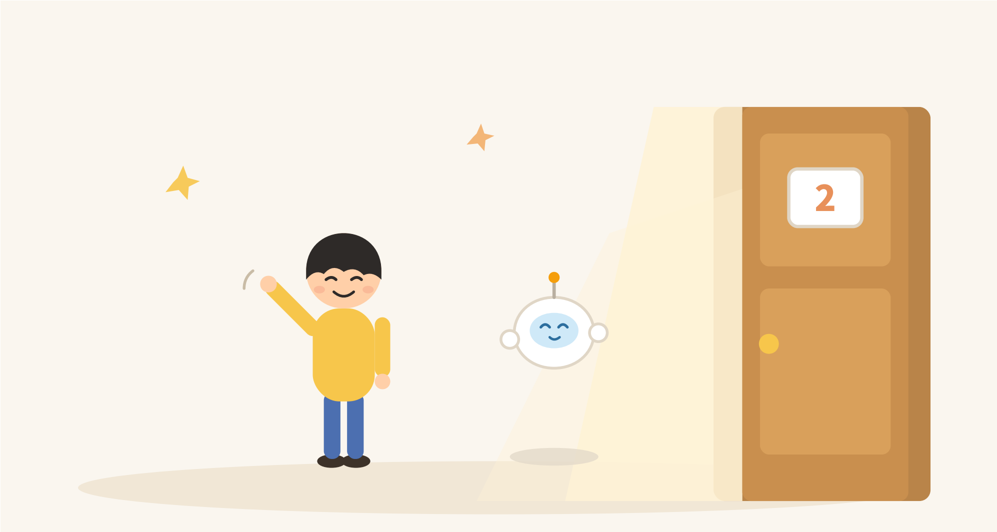
1. **AI has been beside you all along, and all of its ability is patterns learned from data.** (Chapters 1 & 3)
2. **The chatting AI plays a word-chain game at unimaginable scale — that is why it flows, and why it fabricates.** (Chapters 4 & 6)
3. **Hand over what is easy to verify and cheap to get wrong; decide yourself whatever is hard to verify and dear to get wrong — and never leave the driver's seat.** (Chapters 7 & 9)

Try It Yourself: Pack Three Things for the Road

Take out pen and paper and write down the three things you most want AI's help with. Work things, life things, that project you've postponed forever — all fair game.

Written down? Tuck the list inside this book.

See you in Book Two, Putting AI to Work — from "understanding it" to "using it well," where the real fun begins.



(End of Chapter 9 — and of Book One, So That's What AI Is. Book Two, Putting AI to Work: don't be a stranger.)